

Project 5: Circuit Design 2

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EEL3123-C0013: Linear Circuits II

Due Date: 2026-07-28
Last modified: 2026-07-28

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4.A Objective

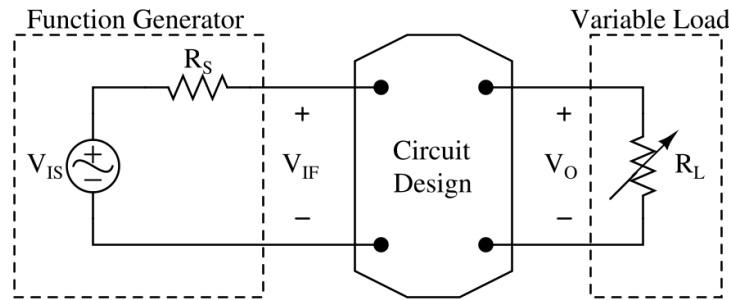


Figure 1: Circuit Model

Design, Analyze, Simulate, Build and Test a two port circuit. Figure 1 shows the desired circuit with a voltage input (V_{IF}) to the left and a voltage output (V_O) to the right.

The voltage input is driven by a function generator. More specifically, an AC voltage source with an internal resistance of 50Ω . The voltage output is caused by a variable load, 600Ω to $30k\Omega$ and the resulting current.

Our objective is to design a circuit that will attenuate a signal less than 200 Hz and pass a signal above 3k Hz. More detailed specifications below (Figure 2)

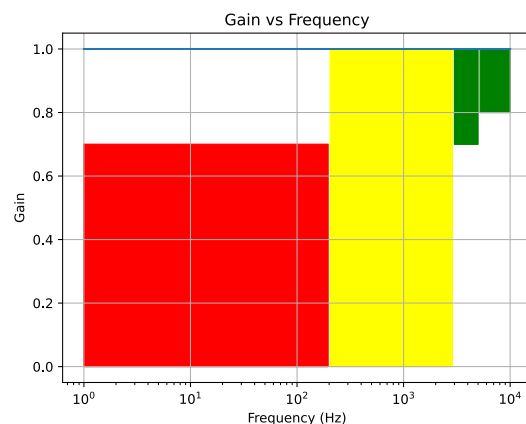
R_{min}	600Ω
R_{max}	$30k\Omega$
$1\text{ Hz} \leq f \leq 200\text{ Hz}$	$ V_O \leq 0.7 \times V_{IS} $
$200\text{ Hz} < f < 3k\text{ Hz}$	$0 \leq V_O \leq 1.0 \times V_{IS} $
$3k\text{ Hz} \leq f < 5k\text{ Hz}$	$ V_O \geq 0.7 \times V_{IS} $
$5k\text{ Hz} < f$	$ V_O \geq 0.8 \times V_{IS} $

Figure 2: Design Specs (Section 13)

f : Frequency
 $|V_{IS}|$: Input Signal Voltage Amplitude
 $|V_O|$: Output Voltage Amplitude

Figure 3 is a bode plot of the frequency bands and their accepted gains for each range. Essentially, if the frequency response graph for all resistances lies entirely in these zones then it is an acceptable circuit.

```
import matplotlib.pyplot as plt
import numpy as np
frequencies = np.logspace(0, 4, 200)
plt.semilogx(frequencies, frequencies / frequencies)
plt.xlabel('Frequency (Hz)')
plt.ylabel('Gain')
plt.title('Gain vs Frequency')
plt.grid(True)
plt.fill_between(frequencies, 0, 0.7, where=(frequencies < 200), color='red',)
plt.fill_between(frequencies, 0, 1.0, where=(frequencies >= 200) & (frequencies <= 3000), color='yellow',)
plt.fill_between(frequencies, 0.7, 1.0, where=(frequencies >= 3000) & (frequencies <= 5000), color='green',)
plt.fill_between(frequencies, 0.8, 1.0, where=(frequencies >= 5000), color='green',)
plt.savefig("Bounds.svg")
```



4.B Plan

Using an initial design of just a series Capacitor, the cutoff frequency changes too much with the load resistance. To stabilize the frequency, we placed a resistor in parallel with the load. This resistor is labeled R_2 in Figure 5.

$$\begin{aligned} R_P &= R_2 \parallel R_L \\ R_{P_{\min}} &= R_2 \parallel 600\Omega \\ R_{P_{\max}} &= R_2 \parallel 30k\Omega \end{aligned} \quad \text{Eq 1}$$

```
import sympy as sp
import matplotlib.pyplot as plt
f, RL, C = sp.symbols('f RL C', positive=True)
Rs = 50
# freq, RL, threshold, sense
conditions = [(200, 30000, 0.7, '<='), (3000, 600, 0.7,
'>=')], (5000, 600, 0.8, '>=')]
print(f"{'R2(ohm)':>8} {'C_min(uF)':>12}
{'C_max(uF)':>12} works?")
for R2 in range(0, 1000, 20):
    Rp = R2 * RL / (R2 + RL)
    g = Rp / sp.sqrt((Rs + Rp)**2 + (1 /
(2*sp.pi*f*C))**2)
    lo, hi, ok = [], [], True
    for fv, RLv, thr, sense in conditions:
        sols = [s for s in sp.solve(sp.Eq(g.subs({f:
fv, RL: RLv}), thr), C) if s.is_real and s > 0]
        if sols:
            (hi if sense == '<=' else
lo).append(float(sols[0]))
        else:
            ok = False
    if not ok or not lo or not hi:
        print(f"{'R2:8d} {'--':>12} {'--':>12} no")
        continue
    C_min, C_max = max(lo), min(hi)
    print(f"{'R2:8d} {'C_min*1e6:12.3f} {'C_max*1e6:12.3f}
{'yes' if C_min < C_max else 'no'}")
```

Analyzing these results subjectively, we felt that the 620Ω resistor would work best for this application. The 620Ω resistor was supposedly available in the lab according to Appendix II of the lab manual.

This gave us a range between $.229\mu F$ and $1.405\mu F$ assuming that all components were manufactured exactly to their value. However, in the real world, this is never the case. To minimize the chance for component tolerances to deviate the circuit behavior from the design specifications, we chose a value as close to the median as possible: $.47\mu F$

R2(ohm)	C_min(uF)	C_max(uF)	works?
0	--	--	no
20	--	--	no
40	--	--	no
60	--	--	no
80	--	--	no
100	--	--	no
120	--	--	no
140	--	--	no
160	--	--	no
180	--	--	no
200	--	--	no
220	--	--	no
240	--	--	no
260	--	--	no
280	--	--	no
300	--	--	no
320	0.947	3.004	yes
340	0.665	2.787	yes
360	0.540	2.601	yes
380	0.465	2.438	yes
400	0.414	2.295	yes
420	0.377	2.169	yes
440	0.348	2.056	yes
460	0.325	1.954	yes
480	0.306	1.862	yes
500	0.290	1.779	yes
520	0.276	1.703	yes
540	0.264	1.634	yes
560	0.254	1.570	yes
580	0.245	1.510	yes
600	0.236	1.456	yes
620	0.229	1.405	yes
640	0.222	1.358	yes
660	0.216	1.314	yes
680	0.211	1.272	yes
700	0.206	1.234	yes
720	0.201	1.197	yes
740	0.197	1.163	yes
760	0.193	1.131	yes
780	0.189	1.100	yes
800	0.186	1.072	yes
820	0.183	1.044	yes
840	0.180	1.018	yes
860	0.177	0.994	yes
880	0.174	0.970	yes
900	0.172	0.948	yes
920	0.169	0.927	yes
940	0.167	0.906	yes
960	0.166	0.887	yes
980	0.164	0.868	yes

Figure 4: Possible values for R_2 and C

However, neither the 620Ω resistor nor the $470nF$ capacitor was available in the lab. We used a 680Ω resistor and three $0.1\mu F$ capacitors in parallel instead.

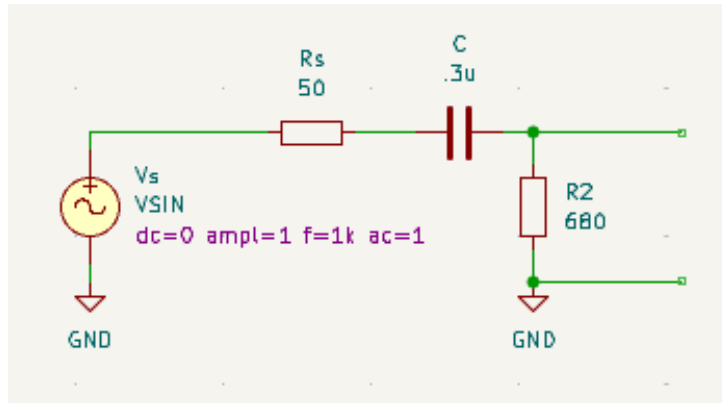


Figure 5: Circuit Design

The following code was used to verify if a Resistor and Capacitor pair would meet the design specifications.

```
import numpy as np
import matplotlib.pyplot as plt
Rs = 50
def gain(f, RL, R2, C):
    Rp = R2 * RL / (R2 + RL)
    return Rp / np.sqrt((Rs + Rp)**2 + (1/(2*np.pi*f*C))**2)
def verify(R2, C):
    checks = [("Stopband 200Hz", 200, 30000, 0.7, '<='),
              ("Passband 3kHz", 3000, 600, 0.7, '>='),
              ("Passband 5kHz", 5000, 600, 0.8, '>=')]
    print(f"R2={R2} ohm, C={C*1e6:.3f} uF")
    all_ok = True
    for name, f, RL, thr, sense in checks:
        g = gain(f, RL, R2, C)
        ok = g <= thr if sense == '<=' else g >= thr
        all_ok &= ok
        print(f" {name:16s} gain={g:.4f} need {sense} {thr} -> {'PASS' if ok else 'FAIL'}")
    print("Overall:", "PASS" if all_ok else "FAIL")
    for RL in range(600, 30000, 1000):
        frequencies = np.logspace(0, 4, 200)
        gains = np.array([gain(f, RL, R2, C) for f in frequencies])
        plt.semilogx(frequencies, gains, label=f'R2={R2} ohm, C={C*1e6:.3f} uF')
    plt.xlabel('Frequency (Hz)')
    plt.ylabel('Gain')
    plt.title('Gain vs Frequency')
    plt.grid(True)
    plt.fill_between(frequencies, 0, 0.7, where=(frequencies < 200), color='red', alpha=0.3, interpolate=False)
    plt.fill_between(frequencies, 0, 1.0, where=(frequencies >= 200) & (frequencies <= 3000), color='yellow', alpha=0.3,
interpolate=False)
    plt.fill_between(frequencies, 0.7, 1.0, where=(frequencies >= 3000) & (frequencies <= 5000), color='green',
alpha=0.3, interpolate=False)
    plt.fill_between(frequencies, 0.8, 1.0, where=(frequencies >= 5000), color='green', alpha=0.3, interpolate=False)
    plt.savefig("Verify-620R-470nF.svg")
verify(R2=620, C=0.47e-6)
# verify(R2=680, C=0.3e-6)
```

Output:
R2=620 ohm, C=0.470 uF
Stopband 200Hz gain=0.3344 need <= 0.7 -> PASS
Passband 3kHz gain=0.8187 need >= 0.7 -> PASS
Passband 5kHz gain=0.8439 need >= 0.8 -> PASS
Overall: PASS

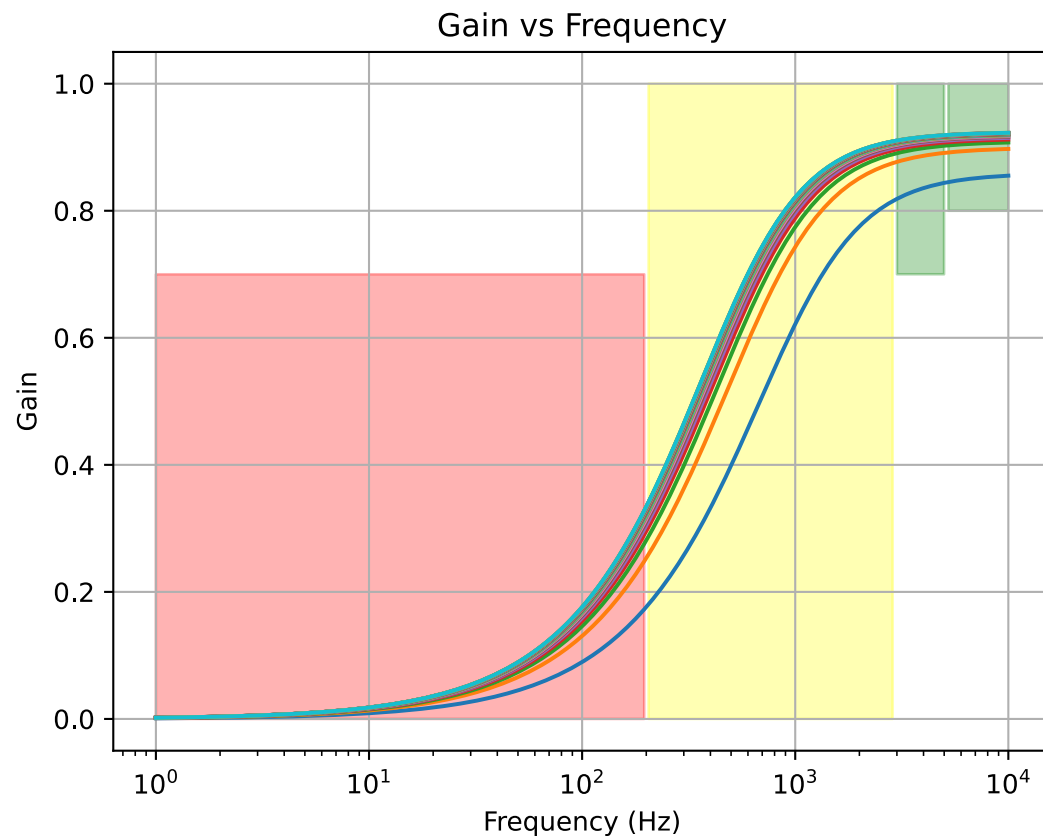


Figure 6: Verify 620R & 470nF

Figure 6 above is the verification of the 620Ω resistor and $470nF$ capacitor. The gain for all three conditions is within the design specifications.

4.C Results

After modifying our design in the experiment phase, we first verified the 680Ω resistor and $300nF$ capacitor in the same manner as the original design.

```
R2=680 ohm, C=0.300 uF
Stopband 200Hz gain=0.2420 need <= 0.7 -> PASS
Passband 3kHz gain=0.7794 need >= 0.7 -> PASS
Passband 5kHz gain=0.8307 need >= 0.8 -> PASS
Overall: PASS
```

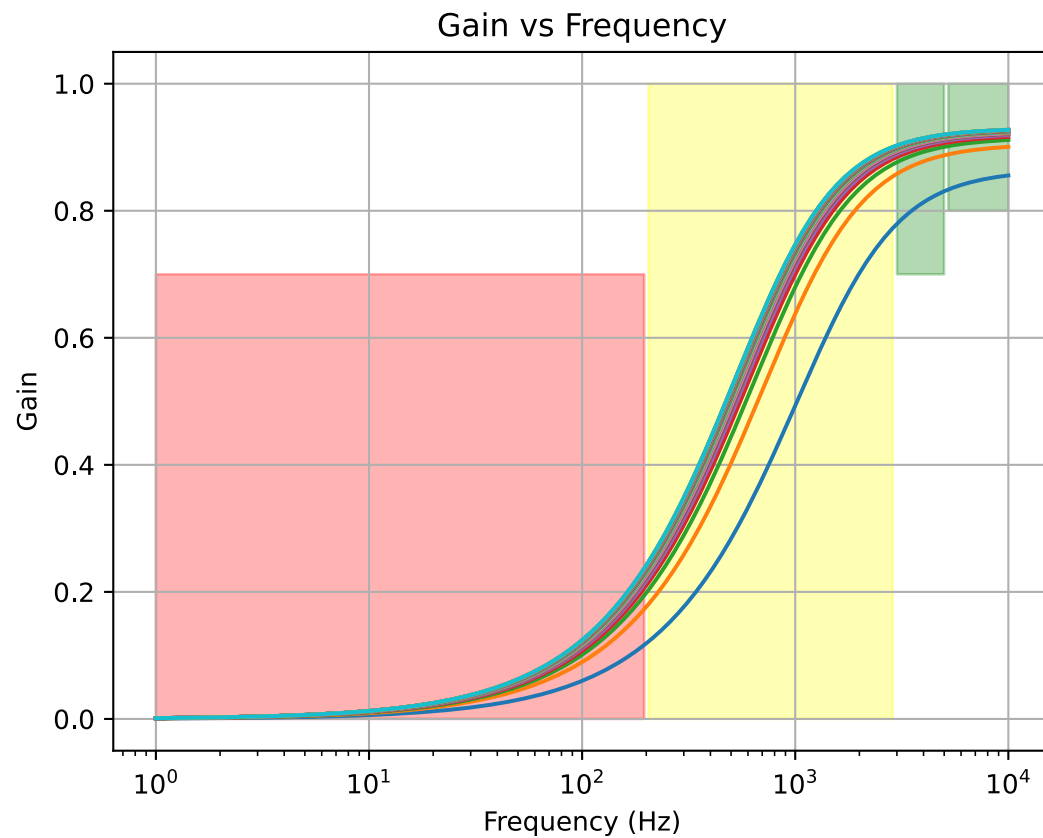


Figure 7: Verify 680R & 300nF

All intermediate values of R_L pass although cutting it a little closer. The gain for all three conditions is within the design specifications.

We simulated the circuit shown in Figure 5. The resulting bode plot is shown below (Figure 8 & Figure 9). The bode plot shows the gain of the circuit for all values of R_L from 600Ω to $30k\Omega$. The gain is within the design specifications for all values of R_L .

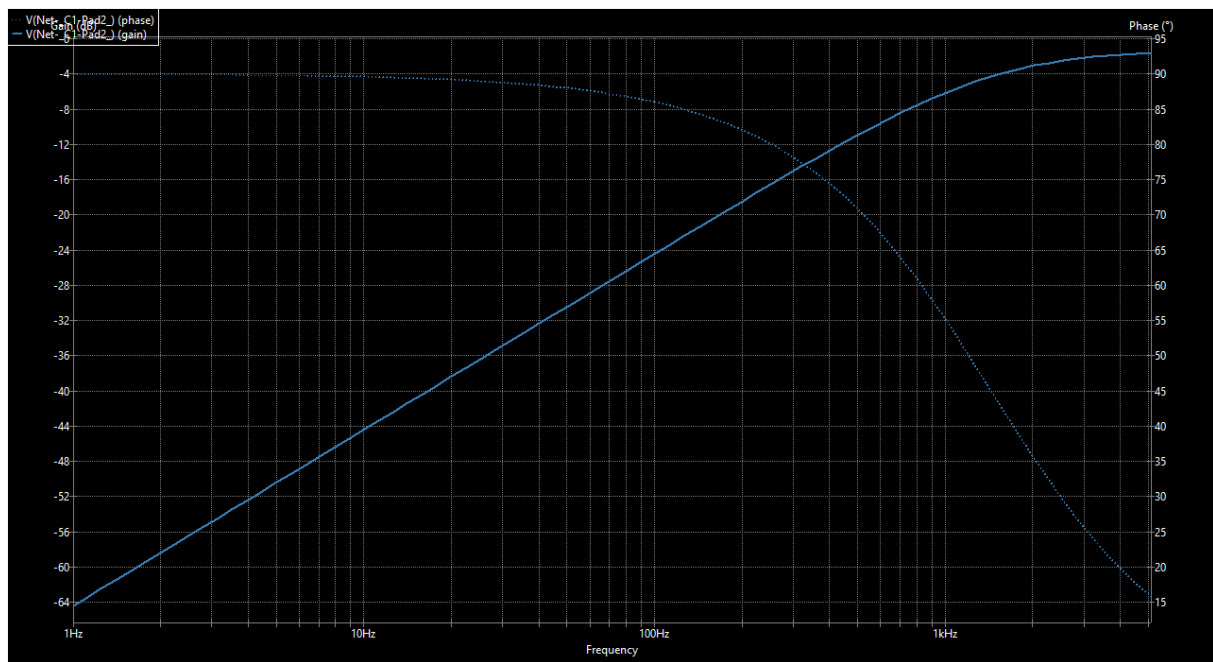


Figure 8: Bode Plot of Simulated Circuit with $R_L = 600\Omega$

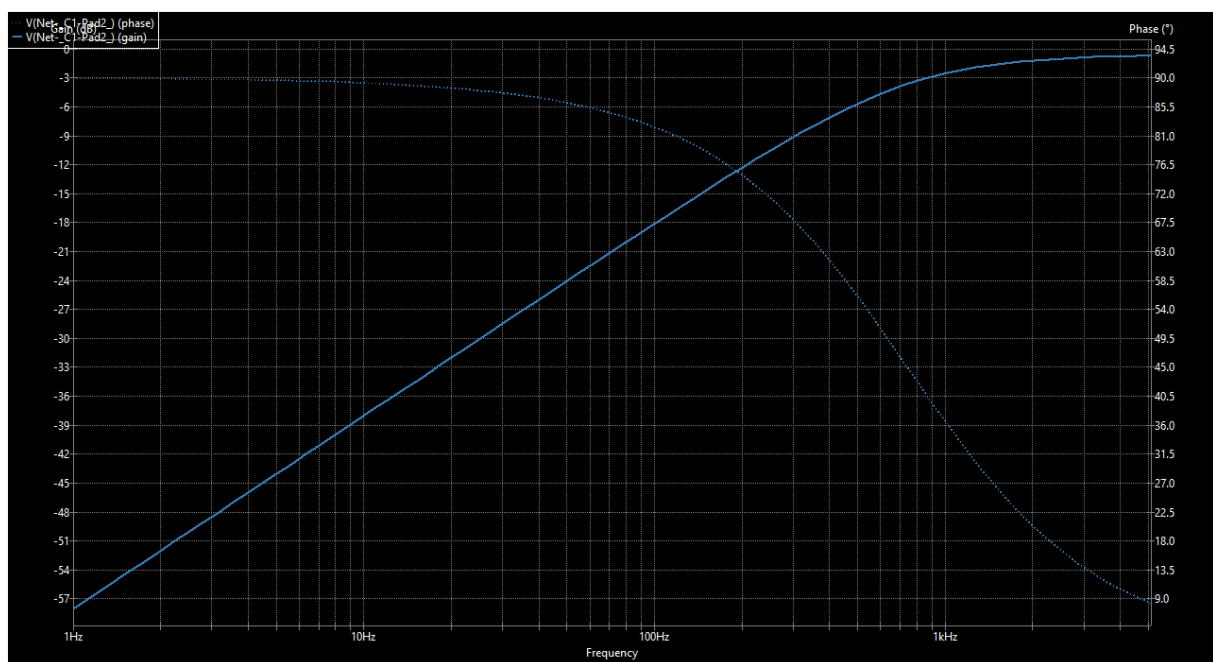


Figure 9: Bode Plot of Simulated Circuit with $R_L = 30k\Omega$

After building the circuit in the lab, we measured the gain at 100 Hz, 4k Hz, and 5.1k Hz for both 600Ω and 30kΩ. The results are shown below (Figure 10).

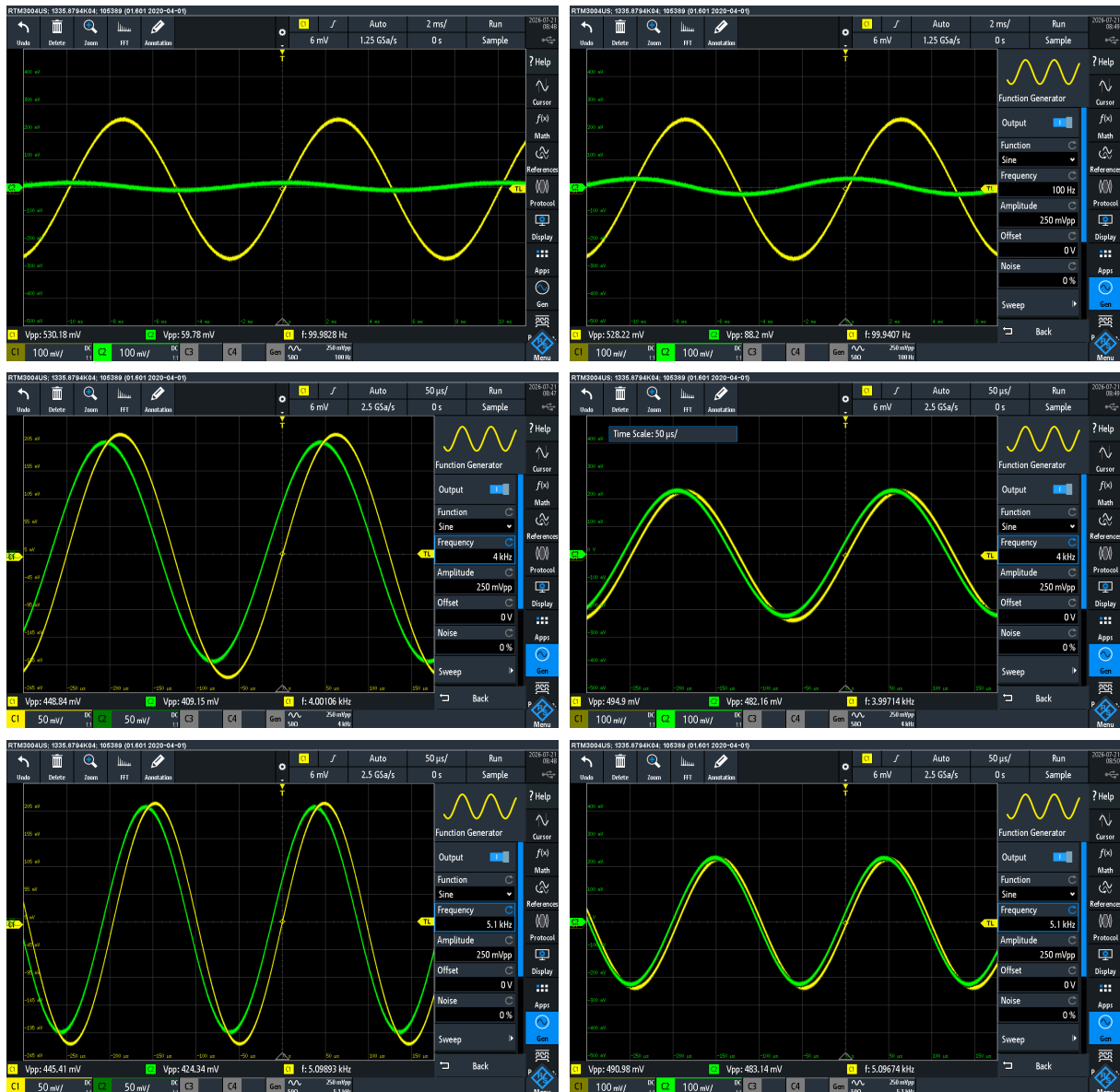


Figure 10: Oscilloscope Gain Results

Left: 600Ω
100 Hz
4k Hz
5.1k Hz

Right: 30kΩ
100 Hz
4k Hz
5.1k Hz

The results of the lab measurements are consistent with the simulation results. The gain at 100 Hz is below 0.7, while the gain at 4k Hz and 5.1k Hz is above 0.7 and 0.8 respectively, for both worst case loads. This demonstrates that the circuit meets the design specifications across the specified frequency ranges.

The bode plots of the entire frequency range for both 600Ω and 30kΩ loads are shown below.



Figure 11: Bode Plot with $R_L = 600\Omega$



Figure 12: Bode Plot with $R_L = 30k\Omega$

4.D Conclusion

The circuit design meets the spec-ed requirements for frequency attenuation and gain across the defined frequency ranges. The combination of a series capacitor and a parallel resistor allowed us to have a relatively stable cutoff frequency throughout the load range. Calculations, simulations and experimental results confirm that the circuit performs as intended.